

FAREHA ASLAM

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Summary

Computer Engineering undergraduate interested in AI infrastructure, machine learning systems, and scalable developer tools. Experienced in building end-to-end applications across multimodal AI, MLOps, local inference, and full-stack development, with a strong focus on performance optimization and real-world deployment.

Experience

Developer Team , Department of Computer Engineering, Jamia millia islamia *Oct 2025 - Present*
Enhancing and maintaining the official department website, adding new features and content updates.

Open Source

gUrrT: Local First Video Intelligence Engine

Open-source package adopted through 5K+ PyPI downloads

Link: <https://pypi.org/project/gurrt/>

Repo: <https://github.com/farehaaslam/gurrt>

- Co-developed gUrrT, an open-source multimodal Video-RAG framework (5K+ PyPI downloads) for semantic video understanding and retrieval.
- Built a retrieval-first indexing pipeline using CLIP embeddings, Faster-Whisper ASR, ChromaDB, reranking, and conversational memory for context-aware video question answering.
- Integrated llama.cpp server with GGUF vision-language models for low-VRAM parallel caption inference, achieving **16X faster** local indexing compared to Ollama-based processing.
- Designed temporal persistence filtering to selectively process persistent scene changes, significantly reducing redundant frame captioning and improving indexing throughput.

Projects

Phishing Website Detection System (End-to-End MLOps Pipeline)

https://github.com/farehaaslam/network_security

Tech Stack: Python, scikit-learn, MLflow, FastAPI, MongoDB, Docker, AWS (S3, ECR, EC2), GitHub Action

- Built a **production-grade end-to-end MLOps pipeline** (ingestion validation transformation training deployment) with **MLflow (DagsHub)** for experiment tracking and model versioning
- Developed a **FastAPI inference service** for real-time and batch predictions; containerized with Docker and deployed via **CI/CD (GitHub Actions AWS ECR EC2)**
- Automated **retraining pipelines (cron jobs)** and managed **model artifacts on AWS S3**

Local voice agent

https://github.com/farehaaslam/voice_agent

Tech Stack: Python, Ollama, LLaMA 3.1, Faster-Whisper

- **Architected a 100% offline, three-stage AI pipeline** (Transcribe Classify Act) capable of executing local file operations and code generation entirely from voice command
- **Implemented structured tool-calling** using LLaMA 3.1 via Ollama to dynamically map user intent to automated file operations and code generation tasks.
- **Designed a full-stack web UI** with Gradio that features live processing states, streaming audio uploads,

Education

B-Tech , computer engineering
Jamia Millia Islamia

Aug 2023 - Present

Skills

Programming Languages: C, C++,Python

AI-ML : PyTorch, Scikit-learn, NumPy, Pandas, LangChain, OpenAI SDK, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Multimodal AI, Semantic Search, Vector Embeddings

MLOps & cloud : MLflow, Docker, GitHub Actions (CI/CD), AWS EC2, AWS S3, Amazon ECR

Inference & Optimization: llama.cpp, Ollama, GGUF Quantization, Faster-Whisper, GPU/Low-VRAM Inference

Database: MongoDB, MySQL, ChromaDB, QdrantDB

Tools and environment: Git ,Github,postman ,Jupyter Notebook,Docker ,ollama, llama.cpp

CORE CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Web Development: Node.js, Express.js, REST APIs, FastAPI

Achievements

solved 500+ data structure and algorithm questions cross platform

<https://codolio.com/profile/fareha>